

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
df=pd.read_csv(r"C:\Users\Reddy Sai Abhiram\Downloads\2. Finance_Analytics_Dataset.csv")
df
```

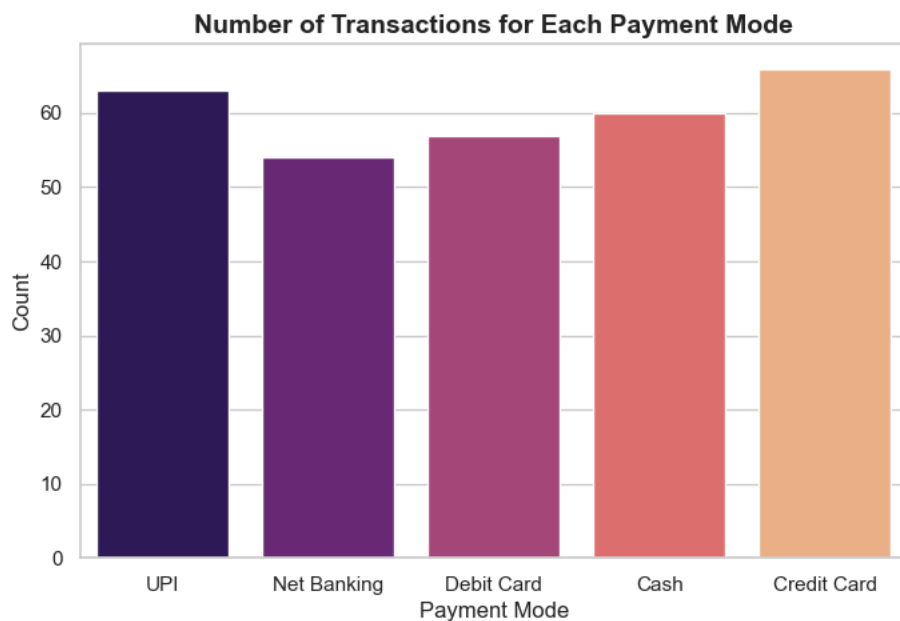
	TransactionID	Date	AccountType	TransactionType	Category	Amount	Balance	PaymentMode	CustomerSegment	FraudFlag
0	1	01-01-2023	Savings	Credit	Groceries	120068	16750	UPI	Retail	Yes
1	2	02-01-2023	Current	Debit	Rent	145359	264540	Net Banking	Retail	No
2	3	03-01-2023	Credit	Debit	EMI	119216	100390	UPI	Corporate	No
3	4	04-01-2023	Savings	Debit	EMI	132038	199151	Debit Card	SME	No
4	5	05-01-2023	Current	Debit	Rent	75523	400540	Cash	SME	No
...	...	...	...	...	...	...	...	...	...	...
295	296	23-10-2023	Credit	Credit	Shopping	139266	50881	Debit Card	Corporate	No
296	297	24-10-2023	Credit	Debit	Rent	118849	303526	UPI	Retail	No

```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='PaymentMode', palette='magma')
plt.title('Number of Transactions for Each Payment Mode', fontsize=14, fontweight='bold')
plt.xlabel('Payment Mode')
plt.ylabel('Count')
plt.show()
```

C:\Users\Reddy Sai Abhiram\AppData\Local\Temp\ipykernel_26428\3606147333.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.countplot(data=df, x='PaymentMode', palette='magma')
```



```
plt.figure(figsize=(6, 5))
sns.countplot(data=df, x='TransactionType', palette='Set2')
plt.title('Number of Transactions by Transaction Type (Debit vs Credit)', fontsize=14, fontweight='bold')
plt.xlabel('Transaction Type')
```

```
plt.ylabel('Count')
plt.show()
```

C:\Users\Reddy Sai Abhiram\AppData\Local\Temp\ipykernel_26428\2693285212.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.countplot(data=df, x='TransactionType', palette='Set2')
```

Number of Transactions by Transaction Type (Debit vs Credit)

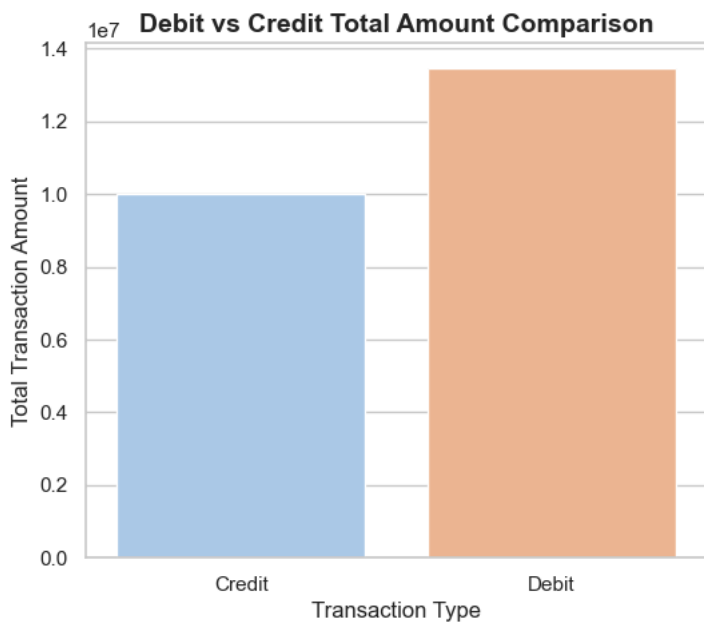


```
plt.figure(figsize=(6, 5))
sns.barplot(data=df, x='TransactionType', y='Amount', estimator=sum, errorbar=None, palette='pastel')
plt.title('Debit vs Credit Total Amount Comparison', fontsize=14, fontweight='bold')
plt.xlabel('Transaction Type')
plt.ylabel('Total Transaction Amount')
plt.show()
```

C:\Users\Reddy Sai Abhiram\AppData\Local\Temp\ipykernel_26428\2420888231.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.barplot(data=df, x='TransactionType', y='Amount', estimator=sum, errorbar=None, palette='pastel')
```

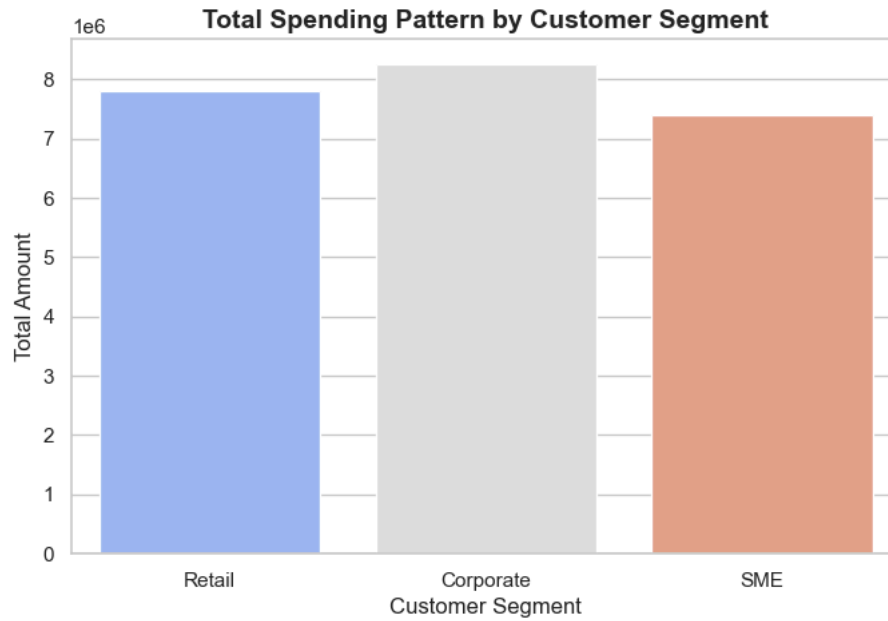


```
plt.figure(figsize=(8, 5))
sns.barplot(data=df, x='CustomerSegment', y='Amount', estimator=sum, errorbar=None, palette='coolwarm')
plt.title('Total Spending Pattern by Customer Segment', fontsize=14, fontweight='bold')
plt.xlabel('Customer Segment')
plt.ylabel('Total Amount')
plt.show()
```

C:\Users\Reddy Sai Abhiram\AppData\Local\Temp\ipykernel_26428\887449175.py:2: FutureWarning:

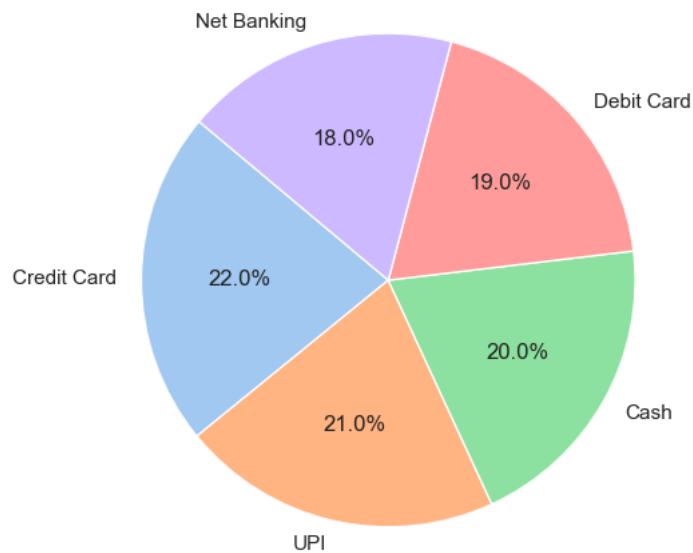
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.barplot(data=df, x='CustomerSegment', y='Amount', estimator=sum, errorbar=None, palette='coolwarm')
```



```
plt.figure(figsize=(6, 6))
payment_counts = df['PaymentMode'].value_counts()
plt.pie(payment_counts, labels=payment_counts.index, autopct='%1.1f%%', startangle=140, colors=sns.color_palette('pastel'))
plt.title('Payment Mode Usage Distribution', fontsize=14, fontweight='bold')
plt.show()
```

Payment Mode Usage Distribution



```
plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='Category', y='Amount', palette='Set3')
plt.title('Boxplot of Transaction Amount by Category', fontsize=14, fontweight='bold')
plt.xlabel('Category')
plt.ylabel('Transaction Amount')
plt.xticks(rotation=45) # Rotates text labels if they crowd together
plt.tight_layout()
plt.show()
```

C:\Users\Reddy Sai Abhiram\AppData\Local\Temp\ipykernel_26428\4261054710.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.boxplot(data=df, x='Category', y='Amount', palette='Set3')
```

